

Lab_Syn_4.0

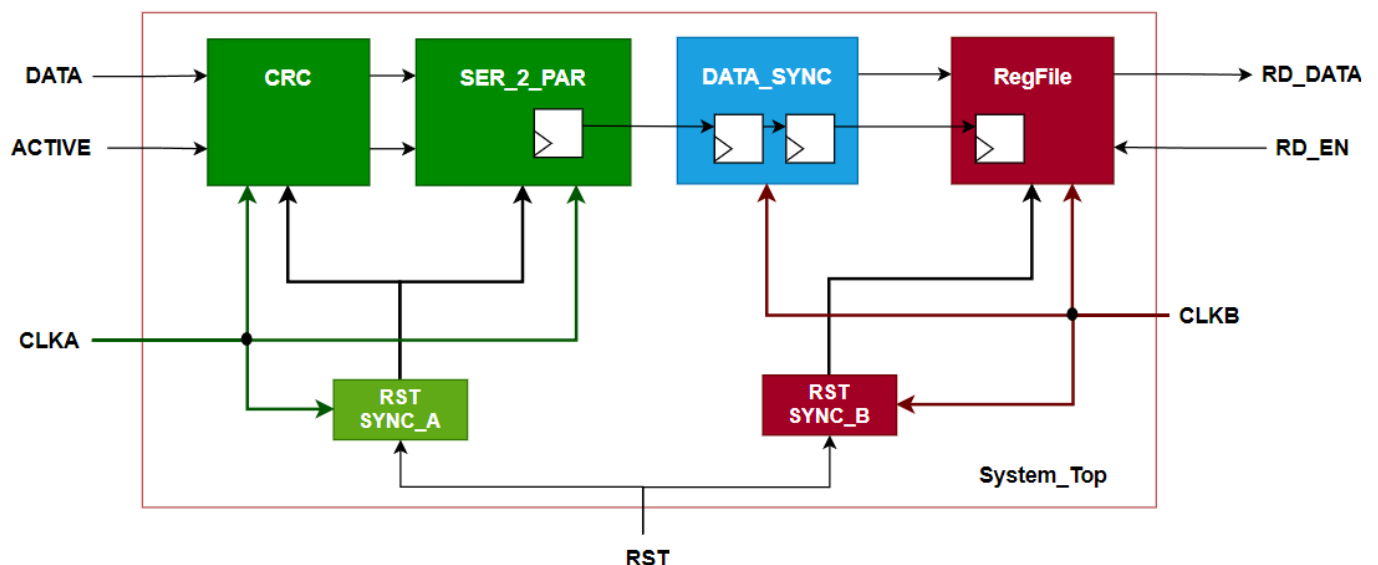
You have to synthesize **System_Top** hierarchy files using **Design Compiler** shell and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std_cells** folder:-

- **scmetro_tsmc_cl013g_rvt_tt_1p2v_25c**
- **scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c**
- **scmetro_tsmc_cl013g_rvt_ss_1p08v_125c**

Steps: -

1. Copy **Lab_Syn_4.0** folder from your computer into the virtual machine inside **Labs** directory.

Here is a simple block diagram for **System_Top** which represent the two asynchronous clocks **CLKA** and **CLKB** and their connections to help you to constraint our system



2. Open a terminal inside path **/home/IC/Labs/Lab_Syn_4.0/syn** and run **syn_script.tcl** script through design compiler shell using the following command in the terminal (**dc_shell -f syn_script.tcl | tee syn.log**)
3. Check the timing paths between **CLKA** and **CLKB**
4. Open **cons.tcl** file and Add the following constraints: -
To exclude the timing paths between **CLKA** domain and **CLKB** domain: -
 - **set_false_path** commands between **CLKA** to **CLKB**
 - OR
 - **set_clock_groups** command between **CLKA** to **CLKB**

5. Open a terminal inside path `/home/IC/Labs/Lab_Syn_4.0/syn` and run again `syn_script.tcl` script through design compiler shell using the following command in the terminal (`dc_shell -f syn_script.tcl | tee syn.log`)
6. Check the timing paths between CLKA and CLKB again. It must be excluded and not included in the timing reports
7. Open `cons.tcl` file: -
 - a) Comment `set_clock_groups` or `set_false_path` commands
 - b) Add the following command
 - `set_max_delay 5 -from [get_cells U0_SER_2_PAR/PAR_DATA*] \`
`-to [get_cells U0_DATA_SYNC/sync_bus*]`